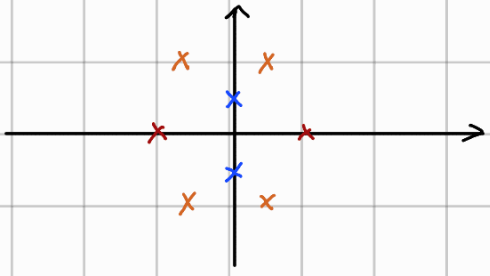
1- Discuss about the following items in the most appropriate and concise way (in English)

- Does the presence of an entire row of zeros in the Routh table always mean that the system has poles on the $j\omega$ axis?
- What are the fundamental causes for the existence of steady state error in a system? (Input signal, transfer function, characteristic equation, type, order, …) discuss about your answer.
- Does the stability of a system changes for various input signals? why?
- How does zero and pole placements affect stability, accuracy and performance (Max. overshoot and settling time) of the system?

a) Not always. When an entire row in Routh table becomes zero, it means that at least two poles are on the imaginary axis or real axis or complex poles symmetric with respect to the origin



b) The stead state error is related with the input of the system and the type of it. For example a type 1 system can follow an step iput while inputs like ramp or parabolic signals are not followe accurately.

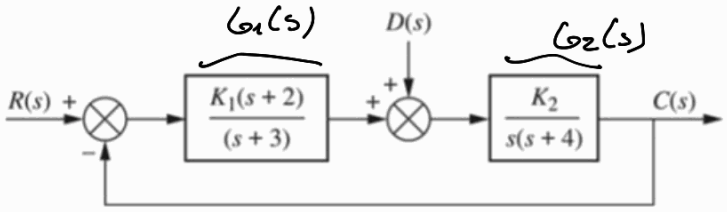
c) No, stability is a characteristic of the system itself and does not depend on the input signal.

d) pole placement:



2- In the given system, determine the two gains K_1 and K_2 so that both following conditions satisfied,

- Steady-state error due to a unit step disturbance is -0.00001
- Steady-state error due to a unit step input is 0.002



Score: 20/100

$$\begin{cases} C = (EG_1 + D)G_2 \\ E = R - C \end{cases} \Rightarrow R - E = C$$

$$R - E = (EG_1 + D)G_2$$

$$R - E = EG_1G_2 + DG_2$$

$$\Rightarrow EG_1G_2 + E = R - DG_2$$

$$E = \frac{R - DG_2}{1 + G_1G_2}$$

$$= \frac{R}{1 + G_1G_2} - \frac{G_2D}{1 + G_2G_2}$$

$$e(\infty) = e_{ss} = \lim_{s \to 0} sE_s = \lim_{s \to 0} \left[\frac{sR}{1 + G_1G_2} - \frac{sG_2D}{1 + G_1G_2} \right]$$

a) $D(s) = \frac{1}{s}$, $R(s) = 0$

$$-0,00001 = \lim_{s \to 0} \frac{-G_2}{1 + G_1G_2} = \lim -\frac{\frac{k_2}{s(s+4)}}{1 + \frac{k_1(s+2)}{s+3}\frac{k_2}{s(s+4)}}$$

$$\Rightarrow 0,00001 = \lim_{s \to 0} \frac{k_2(s+3)}{s(s+3)(s+4) + k_1k_2(s+2)} = \frac{3k_2}{2k_1k_2} = \frac{3}{2k_1}$$

$$\Rightarrow k_1 = 150\,000$$

b) For $R(s) = \frac{1}{s}$, $D(s) = 0 \Rightarrow E_{ss} = 0,002$

$$0,002 = \lim_{s \to 0} \frac{\frac{1}{1 + G_1G_2}}{1 + \frac{k_1(s+2)}{s+3}\frac{k_2}{s(s+4)}} = \lim_{s \to 0} \frac{s(s+3)(s+4)}{s(s+3)(s+4) + k_1k_2(s+2)} \neq 0$$

so the second condition may not be satisfied, and for any value of k_2 the system tracks unit step input!!

3- Determine the range of K in the following transfer function for a typical tape-drive, which leads to a stable system. Note that the closed-loop transfer function is $1+KG(s) = 0$.

$$KG(s) = \frac{K(s+6)}{s[(s+0.7)(s+1.2)(s^2+0.8s+6)]}$$

$1 + k\,G(s) = 0$

$$1 + \frac{k(s+6)}{s[(s+0,7)(s+1,2)(s^2+0,8s+6)]}$$

$$= \frac{s[(s+0,7)(s+1,2)(s^2+0,8s+6)] + k(s+6)}{s[(s+0,7)(s+1,2)(s^2+0,8s+6)]} = 0$$

$s[(s+0,7)(s+1,2)(s^2+0,8s+6)] + k(s+6) = 0$

$\Rightarrow (s^2+0,7s)(s^3+0,8s^2+6s+1,2s^2+0,96s+7,2) + ks + 6k = 0$

$\Rightarrow (s^2+0,7s)(s^3+2s^2+6,96s+7,2) + ks + 6k = 0$

$\Rightarrow s^5 + 2s^4 + 6,96s^3 + 7,2s^2 + 0,7s^4 + 1,4s^3 + 4,872s^2 + 5,04s + ks + 6k = 0$

$\Rightarrow s^5 + 2,7s^4 + 8,36s^3 + 12,072s^2 + (k+5,04)s + 6k = 0$

s^5	1	$8,36$	$k+5,04$
s^4	$2,7$	$12,072$	$6k$
s^3	$\frac{2,7 \cdot 8,36 - 12,072}{2,7} = 3,89$	$\frac{2,7(k+5,04) - 6k}{2,7} = \frac{-3,3k + 13,608}{2,7}$	
s^2	$\frac{(3,89)(12,072) - \frac{(-3,3k+13,608)}{2,7} \cdot 2,7}{3,89} = 33,35 + 3,3k$	$6k$	
s^1	$\frac{k^2 + 28,97k - 161,5}{-8,44 - 0,82k}$		
s^0	$6k$		

in order to have no change in sign, all components of the first column must be positive which gives the following conditions

1) $k > 0$ ← dominant

2) $33,35 + 3,3k > 0 \quad k > -10,1$

$$\boxed{k > 0}$$

Let's check the last condition

$$\frac{k^2 + 28,37k - 41,53}{-8,24 - 0,82k} > 0$$

and k alway $k > 0$

$$\Rightarrow \underbrace{k^2 + 28,37k - 41,53}_{eq} < 0$$

$k < -29,76 \rightarrow eq > 0$ ✗

$-29,76 < k < 1,395 \rightarrow eq < 0$ ✓

$k > 1,395 \rightarrow eq > 0$ ✗

So in total the acceptable range is $0 < k < 1,395$

4- In a unity negative feedback system,

a. Obtain the root locus plot if the open loop transfer function is

$$G(s) = \frac{K(s+1)}{s(s-1)(s^2+6s+30)}$$

b. Determine the range of K for stability, according to the plot.

c. What is the root locus plot for the same system in a positive feedback architecture?

Score: 40/100

Zero : $z = -1$

poles : $p_1 = 0$, $p_2 = 1$, $p_{3,4} = -3 \pm j\sqrt{21}$

unmatched branches : $n - m = 3 \Rightarrow$ 3 asymptotes

$\Phi_k = \frac{2k+1}{n-m} \cdot 180$

$\rightarrow \Phi_0 = 60°$

$\rightarrow \Phi_1 = 180°$

$\rightarrow \Phi_2 = 300° = -60°$

So asymptote angles are $\pm 60°$ and $180°$

The centroid of asymptotes may be determined as

$$s = \frac{\Sigma \text{real part of poles} - \Sigma \text{real part of zeroes}}{n-m} = \frac{-4}{3} = -1,33$$

Now, let's calculate the breaking points by using the characteristic equation

$$s(s-1)(s^2+6s+30) + k(s+1) = 0 \Rightarrow k = -\frac{s(s-1)(s^2+6s+30)}{s+1}$$

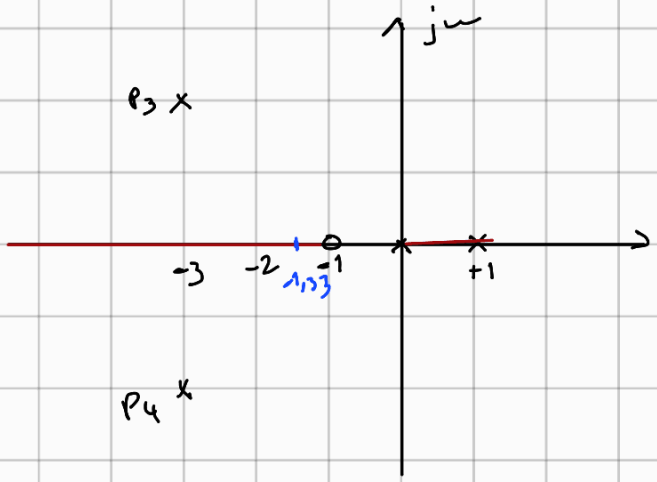
$$\frac{dk}{ds}=0 \Rightarrow \frac{(s-1)(s^2+6s+30)+s(s^2+6s+30)+s(s-1)(2s+6)-s(s-1)(s^2+6s+30)}{(s+1)^2}$$

$$\rightarrow 3s^4+14s^3+39s^2+48s-30=0$$

$s_1 = 0,44$ ✓

$s_2 = -2,601$ ✓

$s_{3,4} = -1,25 \pm 2,67j$ ×



Centroid: $\dfrac{\sum \text{poles } \sigma - \sum \text{zeroes } \sigma}{n-m}$

$n=4$

$m=0$

$$\frac{-3-3+1-(-1)}{3} = \frac{-6+2}{3} = \frac{-4}{3} = -1,33$$

Now lets determine the angle of leaving from the complex pole

* leaving angle : 180° - $\sum$ angles of other poles w.r.t this pole from p_3 + $\sum$ " " zeroes "

$$= 180^\circ - (90^\circ) - \left(180 - \tan^{-1}\left(\frac{\sqrt{21}}{3}\right)\right) - \left(180 - \tan^{-1}\left(\frac{\sqrt{21}}{4}\right)\right) + \left(180 - \tan^{-1}\left(\frac{\sqrt{21}}{2}\right)\right)$$

$$= -50,75^\circ$$

for pole p_4 the leaving angle will be $50,75^\circ$

We can also determine the intersection of the root locus with the imaginary axis by using the routh table as:

$$s(s-1)(s^2+6s+30)+k(s+1)=0$$

$$\hookrightarrow s(s^3+6s^2+30s-s^2-6s-30)+k(s+1)=0$$

$$= s^4+5s^3+24s^2-30s+ks+k=0$$

$$= s^4+5s^3+24s^2+(k-30)s+k=0$$

s^4	1	24	k
s^3	5	$k-30$	
s^2	$30-\frac{k}{5}$	k	
s	$\frac{k^2-155k+4500}{k-150}$		
s	k		

① $k>0$

② $30-\frac{k}{5}>0 \rightarrow k<150$

③ $\frac{k^2-155k+4500}{k-150}>0$

In order to be stable, all the above conditions must hold.

So $k^2-155k+4500<0$ then $38,68<k<116,31$ to make it happen

and finally all the three condition are satisfied for $36,68<k<116,31$

There are two cases where we can construct a full zero row in the s^1 row

for $k=38,68$ and $k=116,31$ this row becomes zero,

The auxilary equation for these two cases are

$\left(30-\frac{k}{5}\right)s^2+k=0$

(a) $k=38,68$: $22,26s^2+38,68=0$

(b) $k=116,31$: $6,74s^2+116,31=0$

so the jw crossings are

$s_{1,2} = \pm j\,1,32$

$s_{3,4} = \pm j\,4,15$

According to the current data, we can plot the root locus to a high degree of accuracy but if we

